Xingyu Zhao

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Education

University of Wisconsin-Madison Sep 2024 - Present
B.S. in Mathematics and Computer Sciences (Honors)
Expected Dec 2026 | GPA: **4.00/4.00**

Fudan University Sep 2022 - Jun 2024
Undergraduate studies in Computer Science
GPA: **3.90/4.00** | Class rank: **1/88**

Research Interests

Learning dynamics and generalization under distribution shift; interpretability and behavioral transfer in language models; multi-agent interaction and its implications for AI safety.

Publications and Manuscripts

[1] **Shattered Compositionality: Counterintuitive Learning Dynamics of Transformers for Arithmetic.**
Xingyu Zhao, Darsh Sharma, Rheeya Uppaal, and Yiqiao Zhong.
NeurIPS 2026, accepted arXiv:2601.22510 | Code

[2] **Cross-Trait Transfer in Subliminal Learning.**
Xingyu Zhao and Yiqiao Zhong.
Under review at ICLR 2027 arXiv:2610.04260 | Code

Research Experience

University of Wisconsin-Madison May 2025 - Present
Undergraduate Researcher, Department of Statistics and Computer Sciences
Advisor: Prof. Yiqiao Zhong

Transformer Learning Dynamics and Compositional Generalization

- Designed controlled experiments across addition, multiplication, comparison, and sorting, identifying counterintuitive orders of subskill acquisition. In the addition task, models learned output digits from the most to least significant place, opposite to the direction of carry propagation.
- Analyzed the conditional-dependence structure of addition data to explain the counterintuitive learning order; developed mutual-information diagnostics to track how model predictions capture training-data dependencies across checkpoints.
- Designed a targeted ablation that randomized only the thousands-place digit of the result; models then failed to learn any lower-place digits, despite unchanged inputs and correct lower-place labels.
- Constructed controlled comparison and sorting tests to isolate subskill interference under distribution shift; showed that conflicting comparison cues impair generalization despite strong in-distribution accuracy.
- Trained transformers with up to 100M parameters from scratch, testing positional encodings and scratchpad supervision; evaluated compositional robustness in 10 pretrained LLMs using mathematical word problems with added reasoning steps.

Research Experience *(continued)*

University of Wisconsin-Madison | Advisor: Prof. Yiqiao Zhong | May 2025 - Present

Cross-Trait Transfer in Subliminal Learning

- Investigated hidden preference transfer between language models through semantically unrelated teacher-generated numbers, symbols, mathematical solutions, and code. Modeled cross-trait effects as directed networks, revealing clusters aligned with color-hue families and geographic groups of cities.
- Developed six diagnostic metrics from output-distribution shifts and internal representations, including activation steering; quantified their Pearson correlations with observed transfer to assess which model signals capture cross-trait structure.
- Applied singular-value analysis to transfer matrices across training checkpoints; increasing effective rank revealed a progression from broad behavioral shifts to more differentiated, trait-specific transfer.
- Designed multi-trait subliminal-transfer experiments using 100 color-preference pairs, each jointly specified in a teacher prompt; opposed latent representations were associated with more uneven transfer than aligned representations, consistent with trait competition.
- Fine-tuned 7–8B checkpoints from Qwen2.5, Qwen3-VL, Llama, Mistral, and OLMo using LoRA; separately compared supervised fine-tuning on teacher-generated outputs with distribution distillation to assess how training objectives affect subliminal transfer.

Selected Coursework

Grades: A in all listed courses.

Advanced mathematics: Analysis I & II; Advanced Linear Algebra; Abstract Algebra; Stochastic Processes; Stochastic Calculus.

The mathematics courses above carry UW-Madison's graduate-coursework designation.

Optimization: Introductory course in linear, integer, and mixed-integer programming.

Theoretical computer science: Algorithms; Data Structures; Theory of Computation.

Machine learning: Deep Learning; Reinforcement Learning.

Systems and graphics: Parallel Programming; Computer Systems; Database Systems; Computer Graphics.

Research Methods and Technical Skills

Model training: Transformer training from scratch; supervised fine-tuning with LoRA; distribution distillation.

Model interpretation: Latent activation analysis; representation geometry; activation steering.

Statistical methods: Mutual-information diagnostics; conditional-dependence analysis; permutation tests with Holm correction; matrix spectral analysis (SVD, effective rank).

Experimental design: Controlled sampling; ablation studies; evaluation under distribution shift.

Scientific computing: Python (PyTorch, Hugging Face Transformers, NumPy, pandas, Matplotlib); C/C++; Java; JavaScript; SQL; Git; Linux.

Selected Software Projects

B-Spline Interactive Demo GitHub

Computer Graphics course project | JavaScript

- Implemented an interactive B-spline visualizer using Cox-de Boor recursion, draggable control points, configurable degree and knot vectors, and real-time piecewise-basis visualization.
- Adopted as an interactive learning tool and lecture aid in Prof. Michael Gleicher's CS559 Computer Graphics course.

Fudan University Study Room Booking Prototype GitHub

Django | SQLite | Bootstrap

- Built a full-stack study-room reservation prototype with server-side booking workflows, persistent relational storage, and a responsive user interface.